

Experimental Validation Plan for a Lightweight Taylor-Cone Solver

Safety-Review Draft

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Approval gate. This document is a proposal for review, not authorization to construct or operate an apparatus. No high-voltage equipment or flammable liquid will be purchased for the testbed, assembled, energized, or tested until the school has approved the final risk assessment, complete schematics, supervision plan, and applicable SRC/ISEF documentation. Any unresolved safety concern stops the physical experiment.

1 Purpose and scientific scope

The project has developed a lightweight two-dimensional axisymmetric electrostatic-capillary solver for Taylor-cone onset. The solver uses an immersed boundary representation, projects the voltage required to balance capillary and Maxwell stresses, and numerically reproduces the classical Taylor-angle region. The purpose of the proposed experiment is to test whether this reduced-order model predicts two directly observable quantities:

1. the cone half-angle and interface profile near onset; and
2. the onset-voltage trend for a fixed liquid and controlled electrode geometry.

The experiment is not intended to demonstrate propulsion, optimize a cone-jet device, or characterize droplet breakup. Its scope is static or near-onset validation of the reduced model. Taylor’s analytical result and later electrohydrodynamic literature provide the theoretical context for the comparison [1, 2, 3, 4, 5].

2 Predictions to be frozen before testing

Before experimental data are viewed, the team will record a frozen simulation package containing the Git commit, dependency versions, geometry, liquid properties, grid resolution, boundary conditions, predicted half-angle, predicted onset voltage, residual landscape, runtime, and numerical uncertainty. Model parameters will not be tuned to force agreement with the final measurements. If a later model revision is necessary, it will be labeled as a separate post-validation analysis.

The primary comparison will report

$$\Delta\alpha = \alpha_{\text{exp}} - \alpha_{\text{sim}}, \quad \delta_V = \frac{V_{\text{exp}} - V_{\text{sim}}}{V_{\text{sim}}} \times 100\%. \quad (1)$$

Agreement will be assessed with uncertainty intervals rather than by requiring an exact 49.29° image measurement.

3 Measured observables and preregistered criteria

3.1 Primary observables

- cone half-angle from calibrated silhouettes;
- interface coordinates and profile root-mean-square error relative to the simulated contour;
- onset voltage under a fixed operational definition;
- repeatability across nominally identical trials;
- electrode spacing, liquid condition, flow setting, temperature, humidity if available, and image metadata.

3.2 Operational onset definition

Before data collection, onset will be defined as the lowest approved voltage setting at which a Taylor-cone-like interface remains within a predeclared geometric stability band for a fixed observation interval. The stability band, observation interval, voltage-step procedure, and rule for ambiguous transitions will be finalized with the supervisor before testing. The definition will not be changed after the final trials are inspected.

4 Apparatus architecture for engineering review

The final engineering package must contain three linked schematics before construction is considered:

1. a complete electrical schematic showing the approved power source, manufacturer-specified current limiting, disconnects, interlock chain, emergency stop, discharge path, measurement instruments, protective earth, and every conductive exposed part;
2. a fluidic schematic showing the liquid reservoir, approved delivery equipment, tubing, fittings, nozzle, containment, spill control, and waste path; and
3. a physical layout showing enclosure dimensions, ventilation path, camera/backlight placement, separation distances, access panels, cables, and operator position.

Students will not choose safety-critical component ratings or certify the earthing arrangement. Those decisions require review by the school and a suitably qualified electrical-safety authority. The final setup must use equipment within its manufacturer-rated operating conditions.

5 Response to high-voltage safety concerns

5.1 Interlock and fail-safe logic

The previously proposed magnetic reed switch is not treated as a complete safety system. The revised design must use an approved, fail-safe interlock architecture whose safety function is reviewed before procurement or construction. The design requirement is that opening any access point causes the source to enter a de-energized state without relying on software or on the reed switch alone. A single component failure must not silently permit energization. Reset must require deliberate supervisor action after the enclosure is closed and the area is checked.

The engineering schematic will document the state sequence for power-off, access-open, access-closed, enabled, tripped, emergency-stop, and reset conditions. The team will not claim that this sequence is safe until it has been reviewed and accepted by the school.

5.2 Residual-energy discharge

Breaking mains input does not prove that the high-voltage output is safe because internal or external capacitance may retain charge. The final design must therefore include a manufacturer-approved passive discharge provision or commercial supply with a documented discharge function. The review package must provide the manufacturer's stored-energy and discharge-time specifications and a supervisor-approved method for verifying a safe state before access.

No student will open or modify the high-voltage supply. A trip will initiate a mandatory wait-and-verify procedure. The enclosure will remain inaccessible until the approved indication and verification procedure confirm that the system is in the safe state. The acceptable voltage threshold, waiting time, measurement method, and responsible person remain unresolved until qualified review.

5.3 Protective earth and bonding

The team will not connect a grounding braid to an improvised laboratory point. Before the experiment proceeds, school facilities staff or a qualified electrician must identify and approve the protective-earth connection available in the intended room. The final drawing must show a single reviewed bonding scheme for the enclosure, supply chassis, extractor, shielding, and other accessible conductive parts, while keeping protective-earth and measurement-return functions unambiguous. Bond continuity and the suitability of the laboratory circuit must be checked by authorized personnel using institutional procedures.

5.4 Current limiting, access, and emergency response

The final apparatus must minimize available current and stored energy using an approved supply and manufacturer-recommended limiting components. All live parts must be inaccessible during operation. The supervisor must control energization, the emergency stop must be reachable without approaching the apparatus, and the operating procedure must define what happens after any trip, unexpected sound, odor, arc, spill, loss of ventilation, or instrument fault. No person will touch, adjust, or retrieve anything from the enclosure while it is energized or awaiting discharge verification.

6 Response to chemical and explosion concerns

6.1 No sealed vapor-accumulating enclosure

The earlier description of a sealed acrylic box is withdrawn. A sealed volume containing ethanol could permit vapor accumulation and is not acceptable without a professional hazardous-atmosphere assessment. The revised design must use a school-approved ventilated location and a ventilation arrangement reviewed for the intended small solvent quantity. The enclosure's purpose is physical access control and secondary containment; it must not be assumed to provide explosion protection.

The final risk assessment must estimate the maximum liquid inventory, maximum credible spill, enclosure volume, ventilation assumptions, and vapor-generation scenario. Testing cannot proceed if the school cannot demonstrate that vapor concentration will remain acceptably below

the applicable limit under both normal and credible fault conditions. Consumer gas sensors or student calculations alone will not be treated as proof of safety.

6.2 Ignition-source control

The high-voltage region is itself a potential ignition source. Consequently, approval requires a documented review of whether ethanol can be permitted at all in the proposed arrangement. All unrelated ignition sources must be excluded from the designated work area, including flames, hot surfaces, switching devices not approved for the location, and spark-producing activities. Only the minimum approved liquid quantity may enter the work area. The final protocol must specify ventilation start and verification, liquid introduction, energization, shutdown, spill response, waste handling, and room-clearance conditions.

If the school concludes that flammable solvent and high voltage cannot be combined safely in its facilities, the physical design must change to an approved lower-hazard validation medium or the project must use external published data. The desire for hands-on validation does not override this gate.

6.3 Chemical controls

The final submission will include the current safety data sheet for every liquid and dopant, approved storage and labeling, compatible spill materials, PPE requirements, maximum quantity, transfer method, waste container, and disposal route. Doping chemistry will not be improvised. No chemical preparation or disposal will occur without supervisor authorization.

7 Optical and fluidic measurement plan

A camera and backlight outside the hazardous-access boundary will record the meniscus silhouette. A calibration target placed in the same imaging plane before testing will establish pixel scale. Exposure, focus, camera position, and magnification will be fixed for a trial block. The image-analysis pipeline will record the selected frames, segmentation settings, fitted flank interval, and uncertainty caused by edge selection and optical resolution.

The liquid-delivery method and flow range must be selected from approved equipment and documented on the fluidic schematic. Because the solver is quasi-static and does not model full cone-jet flow, the primary analysis will use the lowest stable approved flow regime and will not claim general flow-rate prediction.

8 Trial matrix and analysis

The first approved campaign will deliberately remain small:

The analysis will report individual measurements, mean, standard deviation, confidence or uncertainty intervals, excluded trials with reasons, and model-experiment differences. A result that disagrees with the solver will be reported rather than hidden or tuned away.

9 Supervision, timeline, and access request

The physical work is divided into approval, dry commissioning, and data collection. The following is an initial estimate for discussion, not a demand on the supervisor's schedule:

Element	Proposed minimum	Purpose
Baseline condition	One fixed geometry and liquid condition	Establish repeatability before any sweep
Repeated trials	At least five independent onset determinations, subject to approval	Estimate mean, spread, and transition ambiguity
Electrode spacing	Two or three approved values	Test the predicted onset-voltage scaling trend
Image frames	Multiple stable frames per trial	Estimate angle and profile uncertainty
Controls	Calibration image, zero-voltage image, recorded environmental conditions	Detect imaging bias and document conditions

Activity	Estimated supervised time	Gate
Final schematic and risk-review meeting	1–2 hours	Written approval before procurement/construction
Equipment inspection and non-energized assembly review	1–2 hours	Supervisor accepts physical layout
Dry commissioning with no flammable liquid	2 hours	Interlock, shutdown, and safe-state procedures accepted
Pilot session under approved conditions	2 hours	Stop/go review before full trials
Baseline repeated trials	3–4 hours	Data-quality and safety review
Optional spacing sweep	3–4 hours	Only if baseline is repeatable and time permits
Shutdown, inventory, and waste closeout	1 hour	Records completed
Total estimated supervised laboratory time	13–17 hours	Spread across approved sessions

A proposed calendar will be produced only after Mr. George identifies possible supervision windows and the school confirms the approval process. If the required one-to-one supervision cannot be scheduled, the experiment will not proceed. The project can still be completed using numerical verification and peer-reviewed external datasets.

10 Approval gates and stop-work criteria

Physical work may proceed only after all of the following are complete:

- school and applicable SRC/ISEF approval obtained before experimentation;
- final electrical, fluidic, grounding, ventilation, and physical-layout schematics accepted;
- risk assessments and safety data sheets accepted;
- qualified review of protective earth, discharge, current limiting, and interlocks completed;
- supervision hours and room access confirmed in writing;
- training, emergency procedure, and waste route documented;

- dry commissioning completed before any approved liquid is introduced.

Work stops immediately for any arc, unexplained trip, loss of ventilation, spill, damaged cable or enclosure, failed interlock, uncertain grounding, absent supervisor, or deviation from the approved schematic. Students will not troubleshoot an energized system.

11 Open questions for school review

The following points cannot be resolved by the student team alone:

1. Is a high-voltage/flammable-liquid experiment permissible in the school facility at all?
2. Who is qualified to approve the protective-earth and residual-energy controls?
3. What ventilation or hazardous-atmosphere assessment does the school require?
4. Which approved power source, enclosure, interlock components, instruments, and liquid-delivery equipment are available?
5. What maximum liquid inventory and chemical condition may be authorized?
6. Which SRC/ISEF forms and institutional reviews must be completed before work begins?
7. Can the estimated 13–17 hours of one-to-one supervision be scheduled?

12 Required attachments before approval

The next review package will append: complete schematics; bill of materials with manufacturer documentation; supply discharge specifications; institutional earthing review; ventilation assessment; chemical safety data sheets; risk assessment; operating and emergency procedures; training record; supervision calendar; applicable competition forms; and the frozen simulation prediction sheet. Until those attachments are complete and accepted, the experiment remains a proposal.

References

- [1] G. I. Taylor. Disintegration of water drops in an electric field. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 280(1382):383–397, 1964.
- [2] Juan Fernández de la Mora. The fluid dynamics of taylor cones. *Annual Review of Fluid Mechanics*, 39:217–243, 2007.
- [3] J. R. Melcher and G. I. Taylor. Electrohydrodynamics: A review of the role of interfacial shear stresses. *Annual Review of Fluid Mechanics*, 1:111–146, 1969.
- [4] D. A. Saville. Electrohydrodynamics: The taylor–melcher leaky dielectric model. *Annual Review of Fluid Mechanics*, 29:27–64, 1997.
- [5] R. P. A. Hartman, D. J. Brunner, D. M. A. Camelot, J. C. M. Marijnissen, and B. Scarlett. Electrohydrodynamic atomization in the cone-jet mode: Physical modeling of the liquid cone and jet. *Journal of Aerosol Science*, 30(7):823–849, 1999.